

Tensor_operations_flops() : examine operation costs

Flops: floating point operation: addition/multiplication

⇒ US executive order: any foundational model trained
⇒ 100 GPU: ~~312~~ TFLOPS $\geq 1e^{26}$ Flops must be reported to the government.

100 GPU: (312) 1979 TFLOPS

$x \rightarrow [B, D]$

$y \rightarrow [D, W]$

$\Rightarrow x @ y \Rightarrow$

FLOPS: $2 \times B \times D \times W$

[rough estimate: include model

no operations in DH is larger than $Matmul$

FLOPS depends on data type/precision

MFU: model FLOPS utilization. > 0.5 is good!

$\Rightarrow (\text{actual FLOPS}) / (\text{promised FLOPS})$

FLOPS = total FLOPS / total time

$\downarrow$
 ~~$2 \times B \times D \times W$~~
 $2 \times \text{tokens} \times \text{param}$

$\downarrow$ = Matmul-time (tokens, params)

Tokens
model parameters

FLOPS

depends on

$\downarrow$

precision

$\downarrow$

sequence length

torch.tensor()

Summary:

- #1: matrix multiplication: dominate $\Rightarrow (2, m, n, r, \text{Flops})$ data type
mode type
sequence length
- #2: Flop/s depends on hardware (H100 $\gg$ A100)
- #3: Model Flop Utilization (MFU): and data type
(BF16 $\rightarrow$ FP32)
(actual Flop/s) / (promised Flop/s)

Gradient-basics()

↓ tensors: { Data, params } $\Rightarrow$ operation forward $\rightarrow$ compute loss

adjust $\Leftarrow$ optimization $\Leftarrow$ gradients $\Leftarrow$ backward $\Leftarrow$

weights Adam SGD

Gradient-flops() $\rightarrow$ linear model: $x \rightarrow w_1 \rightarrow h_1 \rightarrow w_2 \rightarrow h_2 \rightarrow \text{Loss}$

Count flops for gradient calculation on

B: 11384 Num of points

D: 32768 Num of Dimension

K: 8192 Number of ~~out~~ outputs

flops $\rightarrow 2 * (\# \text{ data points}) * (\# \text{ parameters})$

Num of forward path: $(2 * B * D * D) + (2 * B * D * K)$

Num of backward path: $+ 2 * B * D * K + (2 * B * D * D) \rightarrow = 4 * (\# \text{ data points}) * (\# \text{ params}) = \text{flops}$